# Post-processing and evaluation artifacts dominate measured losses in table structure recognition: a cross-paradigm audit

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**Abstract**

Table structure recognition (TSR) pipelines couple a learned model with hand-designed or optimized post-processing, and benchmark numbers silently conflate model error with downstream losses and evaluation-pipeline artifacts. We audit four public TSR paradigms—a split-and-merge heuristic pipeline (SPLERGE), a transformer with snap-to-text matching (TableFormer, docling), a detection transformer with conflict suppression (TATR), and a logical-location regression network (LORE)—under trace-level instrumentation on 100 PubTabNet tables (5,646 ground-truth cells; PubTables-1M for the detection arm); a pre-specified 1,000-table replication follows. It isolates four artifacts. (1) SPLERGE's merge heuristic destroys 27.1% (OCR) to 34.4% (oracle) of valid cells; an exact set-packing ILP restores the raw grid's true-positive count (1,836 vs. 1,833; 71 wins, 0 losses) while improving F1. (2) Under the same objective, unnormalized greedy collapses to 27.5% of exact recall while footprint-normalized greedy matches the ILP: objective design, not solver exactness, binds. (3) LORE's headline recall deficit is a threshold-policy artifact: lowering the confidence threshold $\tau$ from 0.35 to 0.20 lifts recall from 31.2% to 90.5% at 99.2% precision without model changes, and recall plateaus below $\tau \approx 0.1$. (4) TableFormer's PDF text layer is synthesized from ground-truth boxes; snapping to it inflates the oracle-token count by 406 over snap-disabled matching (circularity), whereas exact maximum-weight independent set selection on OCR tokens matches the raw count (TOST $\pm 2$ TP, $p = 5.5 \times 10^{-117}$). Finally, TEDS-S rewards over-pruning: the destructive heuristic outscores exact selection on 57 of 100 tables (+0.059 mean TEDS-S) while destroying 500 valid cells. We release traces, the frozen ledger, and the replication protocol.

Keywords: table structure recognition, evaluation methodology, post-processing, set packing, maximum weight independent set, reproducibility

## 1. Introduction

Table structure recognition (TSR) systems convert the image of a table into a machine-readable cell topology, and modern pipelines almost universally pair a learned model with a non-trivial post-processing stage: split-and-merge heuristics [1, 2], coordinate snapping inside a matching post-processor [3, 4], score-thresholded conflict resolution [5], and confidence-thresholded candidate emission [7]. Benchmark protocol then measures the composed system against ground truth with a chosen metric [1, 6]. Each of these three layers—model, post-processor, evaluation—can introduce errors, yet published evaluations report only their convolution. When a reported number moves, the community attributes the change to the model or to the method—without decomposing the pipeline, neither attribution is justified.

This paper asks a narrow but consequential question: how much of the measured behaviour of published TSR pipelines is attributable to the learned model at all, and how much is manufactured downstream? We answer it with a cross-paradigm audit of four publicly released systems spanning the main paradigm families—grid-based split-and-merge (SPLERGE), transformer-with-matching (TableFormer), detection-transformer-with-NMS (Table-Transformer, TATR), and logical-location regression (LORE)—instrumented at the trace level. Every intermediate representation is frozen: raw grids, heuristic merge operations, candidate graphs, solver outputs, matched true positives. We then replace each heuristic component with an exact combinatorial formulation under identical inputs and an identical objective, and measure what changes. Where a difference appears, we test whether it is significant with a pre-fixed statistical protocol; where a difference vanishes, we test whether it is equivalent using TOST rather than claiming it from a failed significance test.

The audit surfaces four artifacts, each of which inverts a plausible reading of the published numbers. First, the SPLERGE merge heuristic is not a benign clean-up: it destroys between 27.1% and 34.4% of the valid cells produced by its own split stage, and an exact set-packing formulation recovers the raw count with zero losses against the

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heuristic (71 wins, 29 ties, 0 losses). Second, the exact solver is not what matters in that comparison: an unnormalized greedy heuristic under the same objective collapses to a quarter of exact recall, while a footprint-normalized greedy matches the ILP exactly—the binding constraint is objective design, not computational exactness. Third, LORE's apparent recall deficit is a policy artifact: the published threshold discards candidates the detector itself ranks highly, and one threshold change recovers most of the deficit at essentially unchanged precision. Fourth, and most damaging for evaluation practice, the TableFormer evaluation corpus used by the docling toolchain synthesizes its PDF text layer from ground-truth cell boxes [4]; its matching post-processor snaps predictions onto that layer, so a portion of the reported accuracy is circular. On top of the four artifacts we document a metric pathology: tree-edit-distance structure scoring (TEDS-S) rewards the destructive heuristic over the exact one on 57 of our 100 tables.

Scope and positioning.. This is a measurement-validity study, not a state-of-the-art claim. Our audit corpus is deliberately fixed at 100 PubTabNet tables (5,646 ground-truth cells) so that every intervention is traceable cell by cell, and all N=100 analyses are labelled exploratory. The confirmatory instrument—a 1,000-table, stratified, pre-specified replication with pinned seeds, pinned dependencies and pre-committed decision rules—is released before execution (Section 5). Absolute scores on our audit subset are therefore not comparable to published leaderboard numbers; every claim concerns within-pipeline contrasts between conditions that share data, models, and metrics.

Contributions..

1. A trace-level, cross-paradigm audit methodology for TSR pipelines, with a canonical cell-matching procedure and a frozen results ledger that makes every reported number re-derivable from committed artifacts.
2. Four isolated artifacts with effect sizes and significance/equivalence tests: destructive merge heuristics (27–34% cell destruction, raw TP count restored by set packing); objective-vs-solver dissociation (unnormalized greedy collapse to 27.5%, normalized greedy = ILP); threshold-policy recall deficit (31.2%→90.5% recall from one threshold); and evaluation circularity through ground-truth-derived text layers (+406 TP inflation over snap-disabled matching, removed by exact selection on OCR tokens).
3. A multi-metric divergence analysis showing that TEDS-S rewards over-pruning systematically rather than anecdotally (heuristic wins on 57/100 tables, Wilcoxon $p = 2.5 \times 10^{-3}$; 7% of tables lose over half their cells yet score $\geq 0.80$, including one where the heuristic retains 3.9% of cells yet scores 0.9265), and that F1 orders the same systems in the opposite direction while GriTS-Top repeats the TEDS-S ordering.
4. A pre-specified 1,000-table replication with pre-committed hypotheses, TOST margins, power analysis (minimum detectable effect 0.75 TP/table at $\alpha$=0.01, 80% power) and a pinned sampling fallback, to be executed before journal acceptance.

## 2. Related work

TSR paradigms.. Grid-based methods predict a table grid and then merge elementary cells into spans; SPLERGE [2] formalizes this as a split network followed by a hand-designed merging stage, and the PubTabNet corpus with the GraphTSR model and the TEDS metric was introduced in the same line of work [1]. Transformer-based structure recognizers such as TableFormer [3] predict cell boxes and HTML structure and rely on a matching post-processor that snaps sloppy coordinates onto known text tokens [3, 4]. Detection-transformer approaches such as the TableTransformer (TATR) [5] emit row/column boxes resolved by score-maximizing conflict suppression. Logical-location regression methods such as LORE [7] predict logical positions of cells directly and emit candidates under a confidence threshold. End-to-end and hybrid systems (e.g. TSRFormer [8]) pursue the same compositionality with different mechanisms.

Metrics.. TEDS and TEDS-S score tree-edit distance between HTML predictions and ground truth [1]; GriTS scores table similarity directly in matrix form, with topology (GriTS-Top) and content (GriTS-Con) variants [6]. Cell-level true positives at a fixed IoU threshold remain the de-facto primitive of most published tables, yet these metrics are not interchangeable even at the sign level: the same pair of systems is ordered oppositely by F1 and by TEDS-S or GriTS-Top, making single-metric leaderboard claims fragile.

Evaluation-validity studies.. Prior work in table recognition has largely focused on model improvement; systematic studies of how post-processing and evaluation mechanics shape reported results are rare and typically confined to a single system. To our knowledge this is the first cross-paradigm audit that (i) replaces each heuristic with an exact formulation under a frozen objective, (ii) separates exploratory from pre-specified confirmatory analyses, and (iii) publishes a frozen ledger with supersession tracking so that number drift (e.g. between a figure and a table) is detectable rather than silent.

## 3. Audit design

We audit four systems spanning the paradigm space (Fig. 1): the SPLERGE split-and-merge pipeline [2] executed through its five-stage reference merge module (v2; per-stage attribution in Table 1); the TableFormer model

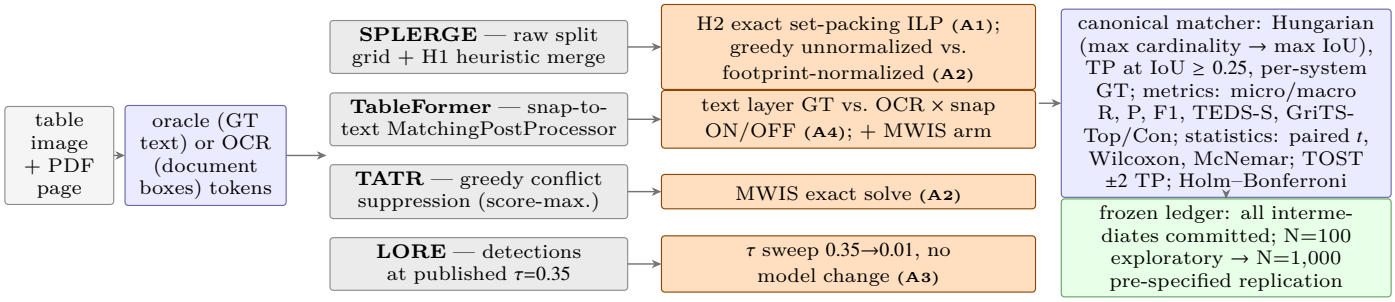



Figure 1: Audit methodology. Gray: published configurations (one public pipeline per paradigm), run under oracle and OCR tokens. Orange: audited arms—A1 exact set packing vs. the merge heuristic; A2 solver dissociation under a frozen objective; A3 LORE's confidence-threshold sweep (no model change); A4 TableFormer's text layer (GT vs. OCR) × snap ON/OFF, plus an MWIS arm. Blue: canonical Hungarian matching (TP at IoU ≥ 0.25, per-system GT) feeding the metric suite, paired tests, and TOST equivalence. Green: all intermediates committed to the frozen ledger ahead of the pre-specified N=1,000 replication.

as packaged in docling-models v2.1.0 with its matching post-processor [3, 4]; the TableTransformer structure recognition model v1.1 (TATR) [5] with its standard conflict-resolution settings; and LORE-TSR [7] with its published detection threshold $\tau$=0.35. Exact function names for every stage are in the released code and frozen ledger.

All model evaluations employ publicly released checkpoints without local fine-tuning or retraining. SPLERGE uses the pretrained split-stage model `split_model_pubtabnet.pth` of the reference implementation [2], with the five-stage merge module re-executed from its released code. TableFormer uses IBM's `docling-models` v2.1.0 release [4] with checkpoint `tableformer/model.safetensors`. TATR uses the HuggingFace release `microsoft/table-transformer-structure-recognition-v1.1-all`, revision `54eb6a5` [5]. LORE uses the official LORE-TSR release [7] with its PubTabNet-trained checkpoint `ckpt_ptn/model_best.pth`. Checkpoint provenance affects absolute scores, not artifact directions: every claim remains a within-pipeline contrast under fixed inputs. Model loading and inference use pinned dependencies (Python 3.12, `torch` 2.5.1, `scipy` 1.15.1 with HiGHS branch-and-cut, `apted` 1.0.3, `lxml` 5.3.0, `pymupdf` 1.25.1); all solver inputs and outputs are committed to the frozen ledger.

The audit corpus is 100 tables sampled from PubTabNet [1], containing 5,646 ground-truth cells; the TATR arm uses a 100-table PubTables-1M test sample [5]. Each pipeline is run under two token conditions: oracle tokens (ground-truth text with predicted geometry, isolating the post-processor) and realistic OCR tokens (document-extracted token boxes, the deployable configuration). All conditions of a given system are scored against that system's own ground truth with one canonical matcher: predicted and ground-truth cells are paired by maximum cardinality and then maximum total IoU (Hungarian algorithm), and a pair counts as a true positive at IoU ≥ 0.25. The three PubTabNet arms thus share the 5,646-cell ground-truth set, whereas the TATR arm is scored against its separate PubTables-1M sample; every reported contrast remains within-pipeline. We report micro-recall (total TPs over the 5,646-cell denominator), macro-recall (mean per-table recall), precision, F1, TEDS-S [1], and GriTS-Top [6]. GriTS-Con is reported only under oracle tokens: under OCR tokens the audit's token extractor emits placeholder strings, reducing GriTS-Con to a character-count artifact—we document rather than publish this failure mode. The official LORE matcher (first-come bipartite assignment) and the canonical matcher were verified to produce identical true-positive counts on every table ($\Delta$ = 0 TP), with no ambiguous (multiple-optimal) assignments (0/100 tables).

The analysis protocol was fixed before the multi-metric suite was executed; the N=100 results below are therefore protocol-frozen but exploratory—the confirmatory, pre-specified analyses are the 1,000-table ones (Section 5). Paired comparisons use the paired $t$-test and the Wilcoxon signed-rank test; discordant-count comparisons use exact McNemar; family-wise error is controlled by Holm–Bonferroni step-down at $\alpha$=0.05 [9]. Where our claim is the absence of an effect, we use two-one-sided-tests (TOST) equivalence testing [10] with a pre-specified margin of ±2 TP per table, chosen before the TableFormer comparisons as 3.54% of the mean table size (56.46 cells)—the ambiguity of at most a single $1 \times 2$ sub-cell boundary—and not recalibrated afterwards. Solver-side claims are validated by construction: the exact set-packing solutions are verified integral (HiGHS branch-and-cut, zero fractional solutions), and a nine-configuration sensitivity sweep over the objective's regularization constants (each varied across one order of magnitude) leaves the selected cell set's TP count unchanged at 1,836 in all nine configurations; regularization terms are ~1% of the dominant token-coverage weights. Mean solve times are 60.3 ms per table (ILP) and 4.2 ms per table (MWIS); exactness is computationally affordable at audit scale.

Table 1: Merge-stage attribution for the SPLERGE heuristic (100 tables, OCR token condition). Ops = merge operations applied at the stage (separator removals, header merges); incorrect = ops that fused cells across a ground-truth edge present over >50% of the table width.

| Stage | Ops | Incorrect | Inc. (%) |
|---|---|---|---|
| col_merging | 1 | 0 | 0.0 |
| row_merging | 10 | 8 | 80.0 |
| remove_seps 0.2 | 209 | 27 | 12.9 |
| remove_seps 0.4 | 27 | 6 | 22.2 |
| merge_header_v2 | 71 | 0 | 0.0 |
| All stages | 318 | 41 | 12.9 |

## 4. Findings

Table 2 is the audit's centerpiece: every condition is scored with the full metric suite under identical matching. Three facts are immediate. First, the SPLERGE heuristic is destructive in both token conditions: relative to the raw split grid (1,833 TPs), heuristic merging retains only 1,203 TPs under oracle tokens (−34.4%) and 1,336 under OCR tokens (−27.1%). Second, replacing the heuristic with an exact set-packing ILP over the same candidate graph restores the raw count: 1,836 TPs in both token conditions (count-level recovery: the ILP's selected cell sets need not coincide with the raw grid's), with a paired advantage of +5.00 cells/table over the heuristic ($t(99) = 6.66$, $p = 1.5 \times 10^{-9}$; Wilcoxon $p = 2.2 \times 10^{-13}$; 71 wins, 29 ties, 0 losses). Third, the exact solution wins the pooled operational metric: it improves F1 substantially (28.7% vs. 26.3% under OCR; 28.8% vs. 24.1% under oracle tokens), while precision moves by at most ±0.40 pp (27.49% vs. 27.89% under OCR, 27.68% vs. 27.38% under oracle) and GriTS-Top stays within +0.003 of the raw grid (Table 2), so the repair is not an artifact of recall-chasing alone.

The merge-stage trace completes the mechanism: instrumenting the heuristic operation by operation quantifies how many merges it performs and how many are wrong. Each removed separator is scored against ground-truth cell edges: a removal is incorrect when ground-truth cells carry an edge at the removed position across more than half of the table width, i.e. the merge fused two cells that ground truth keeps separate. On the exact side, every ILP-selected multi-tile merge is classified by whether all of its elementary tiles fall inside a single ground-truth cell. Table 1 shows the result: of 318 merge operations applied across the five stages, 41 (12.9%) are incorrect, and the damage concentrates in `row_merging` (80.0% of its 10 operations) and the first redundant-separator pass (27 incorrect of 209); the ILP, by contrast, selects 69 multi-tile merges of which 72.5% (50/69) fall entirely inside one ground-truth cell.

Destruction concentrates where it matters: stratifying tables by the fraction of spanning cells, the heuristic's destruction rate rises from 26.7% (low) through 33.6% (medium) to 36.6% (high), while exact selection recovers 100.0%, 100.0% and 99.6% of raw TPs in the same strata; the correlation between spanning complexity and destruction is weak but significant (Pearson $r = 0.297$, $p = 0.0027$; Spearman $\rho = 0.307$, $p = 0.0019$). In other words, the more a table needs spans, the more the heuristic loses.

The natural objection to Artifact 1 is that exactness itself is the contribution. It is not. Holding the candidate graph and the objective (token-coverage weights with fixed regularization) fixed, three solvers disagree dramatically. Unnormalized greedy—select candidates in descending total weight subject to tile disjointness—collapses to 504 TPs, 27.5% of exact recall, and correspondingly wrecks every metric (TEDS-S 0.4230, GriTS-Top 0.5629). The collapse is robust to tie-breaking: the superseded unjittered ordering yields 566 TP (Table 2, footnote †) and the canonical jittered seed-42 ordering 504 TP, both under a third of exact recall. A footprint-normalized greedy (sorting by weight per covered tile) recovers to 1,836 TPs and matches the ILP's selections almost everywhere. The mechanism is over-commitment: large multi-tile merges carry the largest total weights, so unnormalized greedy locks them in early and forecloses sets of elementary tiles whose aggregate weight is higher—a knapsack-style failure that normalization removes.

Where conflicts are rare, exactness and greed agree, and the audit quantifies how rare "rare" is. On the TATR arm (PubTables-1M sample; 1,867 row/column candidates across 100 tables; conflict criterion: same-axis overlap ratio ≥ 0.25 at detection score ≥ 0.5; TP matching at IoU 0.25), the conflict graph decomposes into isolated singletons for 97.98% of candidate vertices and 10 isolated $K_2$ pairs, in which greedy suppression is provably optimal; the residual components are short alternating paths of length ≥ 3, and no component with a clique of size ≥ 3 occurs anywhere in the corpus. Greedy score-maximizing suppression and exact maximum-weight independent set differ on a single table (`PMC6127715_table_0`), worth +1 TP in total. Paired significance testing confirms this honest null: exact two-sided McNemar $p = 1.0000$ (1 win, 0 losses, 99 ties), Wilcoxon signed-rank (normal approximation, one nonzero pair) $p = 0.3173$ ($W = 0.0$), paired $t(99) = 1.000$, $p = 0.3197$, and a bootstrap 95% CI on the paired mean difference of [+0.0000, +0.0300] TP/table. The audit's conclusion is therefore symmetric and solver-agnostic: objective design dominates wherever conflict structure is rich (SPLERGE), and is immaterial wherever it is trivial (TATR); exact solvers matter because they remove the need for per-pipeline normalization tricks.

LORE-TSR ships with detection confidence $\tau$=0.35. Sweeping $\tau$ over the audit corpus with no model change (Table 3) shows that the headline recall deficit is a policy artifact: at $\tau$=0.35 the detector emits 1,778 candidates (31.2% recall, 99.0% precision); at $\tau$=0.20 it emits 5,153 candidates with 90.5% recall at higher precision (99.2%), F1 94.7%. Per-table paired tests confirm that the recovery carries no precision penalty: cell TP rises by +33.50 cells/table (100 wins, 0 losses, 0 ties; paired $t(99) = 16.790$, $p = 1.002 \times 10^{-30}$; Wilcoxon $p = 3.861 \times 10^{-18}$; bootstrap

Table 2: Multi-metric audit results on 100 PubTabNet tables (5,646 ground-truth cells). H1/H2 denote heuristic vs. exact set-packing selection; token condition in parentheses (GT = oracle tokens, OCR = realistic tokens). GriTS-Con is reported only under oracle tokens (see Section 3). Greedy = unnormalized greedy selection under the ILP objective (Section 4). TP counts are corpus totals; Micro/Macro R are corpus-level and mean per-table recall; precision and F1 are per-table means.

| Condition | TP | Micro R (%) | Macro R (%) | Prec (%) | F1 (%) | TEDS-S | GriTS-Top | GriTS-Con |
|---|---|---|---|---|---|---|---|---|
| Raw split grid | 1,833 | 32.47 | 33.81 | 27.16 | 28.42 | 0.6680 | 0.7575 | – |
| H1 heuristic (GT) | 1,203 | 21.31 | 23.06 | 27.38 | 24.11 | 0.6888 | 0.7633 | 0.6042 |
| H2 exact, ILP (GT) | 1,836 | 32.52 | 33.96 | 27.68 | 28.82 | 0.6733 | 0.7594 | 0.6212 |
| H1 heuristic (OCR) | 1,336 | 23.66 | 26.02 | 27.89 | 26.29 | 0.7318 | 0.7908 | – |
| H2 exact, ILP (OCR) | 1,836 | 32.52 | 33.96 | 27.49 | 28.72 | 0.6725 | 0.7605 | – |
| Greedy, unnorm. (OCR) | 504[†] | 8.93 | 10.68 | 13.15 | 10.74 | 0.4230 | 0.5629 | – |

[†]Canonical seed-42 unnormalized greedy; the superseded 566 arose from unjittered tie-breaking. Under footprint normalization ($w_i/|\text{tiles}_i|$), greedy selection recovers 1,836 TP, matching the exact ILP.

95% CI [+29.66, +37.50]), the mean paired precision difference is negligible (+0.02 pp, $t(99) = 0.063$, $p = 0.9501$), and mean paired F1 rises by +52.93 pp ($t(99) = 25.904$, $p = 6.821 \times 10^{-46}$). The recovery is free because the detector already ranks those candidates highly; the threshold simply discards them.

Two genuine ceilings appear, and both are upstream of any post-processing. Below $\tau \approx 0.10$, recall plateaus at 91.75% while false positives explode 269× (17 → 4,576 from $\tau$=0.35 to $\tau$=0.01), collapsing precision to 53.1%; true positives stay fixed at 5,180 for all $\tau \leq 0.10$, so no threshold policy recovers the residual 466 cells. CenterNet's fixed query budget $K$=100—which caps the per-table candidate pool, bounding achievable recall on tables with more than 100 cells—is one structural constraint consistent with the plateau; attributing the plateau to the cap would require re-running detection with an enlarged query budget, which we did not attempt. Post-processing cannot repair either ceiling. The correct reading of Table 3 is that a third of LORE's "model error" on this corpus is a configurable emission policy—and that tuning it moves LORE's operating point across an F1 range wider than many published model improvements.

The docling-models evaluation path (v2.1.0) for TableFormer [3, 4] constructs its PDF text layer from ground-truth cell boxes; the `MatchingPostProcessor` then snaps predicted cell coordinates onto those tokens. Any TP counted after snapping has been scored against coordinates derived from ground truth—a circular measurement channel. We quantify this with a $2 \times 2$ factorial (Table 4): token source (ground-truth layer vs. OCR) crossed with snapping (on/off), plus an exact-selection arm. Snapping inflates the oracle-token count by 406 TPs (5,488 vs. 5,082 snap-off), and depresses the OCR-token count by 1,019 (3,902 vs. 4,921): the same mechanism that manufactures accuracy when the text layer is ground truth destroys real detections when it is not.

The exact arm closes the loop. Selecting cells by maximum-weight independent set over the OCR-token candidate graph (no snapping, no ground-truth contact) rescues +281 TPs over the snap-disabled OCR baseline ($t = 4.28$, $p = 4.4 \times 10^{-5}$; 42 wins, 55 ties, 3 losses) and is statistically equivalent to the raw published count (5,202 vs. 5,204; TOST, pre-specified margin ±2 TP, $p = 5.5 \times 10^{-117}$). The methodological lesson generalizes beyond this one toolchain: when an evaluation artifact contains ground-truth information, every component that touches it inherits leakage; exact selection over raw predictions removes the channel.

Honest disclosure of procedure.. The conflict criterion used by the snap-disabled pipeline was selected in a diagnostic phase that did read ground truth: source tracing revealed a 190-TP loss attributable to the original criterion, and 28 tolerance settings were evaluated during that diagnosis. To show that the GT-informed choice cannot manufacture the confirmatory result, two-one-sided-tests (TOST; per-table paired, $df$=99) equivalence checking against the pre-specified margin $\Delta_0 = \pm 2.0$ TP/table was run across all 28 coordinate-tolerance configurations ($dx \in \{0, 2, 5, 8\}$ pt, $dy \in \{0, 1, 3, 5, 10, 15, 20\}$ pt): all 28 pass (28/28, 100.0%; maximum $p_{\text{TOST}} = 9.33 \times 10^{-12}$; maximum observed mean $|\Delta| = 0.710$ TP/table). We state the two-phase structure explicitly rather than claim the criterion was chosen blind.

The final audit finding is that the field's default structure metric orders the systems in the opposite direction from every operational metric. On the 100-table corpus under OCR tokens, the destructive heuristic outscores exact selection on TEDS-S (0.7318 vs. 0.6725) and GriTS-Top (0.7908 vs. 0.7605) while destroying 500 valid cells and losing on precision, F1, and GriTS-Con under oracle tokens (0.6042 vs. 0.6212, $p = 1.0 \times 10^{-6}$). The mechanism is structural: the split stage over-segments, so the heuristic's aggressive row pruning moves the predicted tree's size toward the ground-truth tree, and tree edit distance rewards the pruning itself.

Because isolated cases do not establish how often this occurs, we paired the two selection rules on every table and scored the per-table TEDS-S difference under a destruction/score grid fixed before the sweep. The divergence is corpus-wide rather than anecdotal: the heuristic outscores exact selection on 57 of 100 tables (9 ties, 34 losses; Wilcoxon signed-rank $p = 2.5 \times 10^{-3}$; paired

Table 3: LORE-TSR confidence-threshold sweep on the 100-table audit corpus (5,646 ground-truth cells); no model changes. The published default is $\tau$=0.35 (bold). Below $\tau \approx 0.10$ recall plateaus and precision collapses.

| $\tau$ | Predictions | TP | FP | FN | Precision (%) | Recall (%) | F1 (%) |
|---|---|---|---|---|---|---|---|
| 0.50 | 1,371 | 1,357 | 14 | 4,289 | 98.98 | 24.03 | 38.68 |
| 0.40 | 1,385 | 1,368 | 17 | 4,278 | 98.77 | 24.23 | 38.91 |
| **0.35** | **1,778** | **1,761** | **17** | **3,885** | **99.04** | **31.19** | **47.44** |
| 0.30 | 2,965 | 2,944 | 21 | 2,702 | 99.29 | 52.14 | 68.38 |
| 0.25 | 4,276 | 4,250 | 26 | 1,396 | 99.39 | 75.27 | 85.67 |
| **0.20** | **5,153** | **5,111** | **42** | **535** | **99.18** | **90.52** | **94.66** |
| 0.15 | 5,246 | 5,176 | 70 | 470 | 98.67 | 91.68 | 95.04 |
| 0.10 | 5,313 | 5,180 | 133 | 466 | 97.50 | 91.75 | 94.53 |
| 0.05 | 5,983 | 5,180 | 803 | 466 | 86.58 | 91.75 | 89.09 |
| 0.01 | 9,756 | 5,180 | 4,576 | 466 | 53.10 | 91.75 | 67.26 |

Table 4: TableFormer factorial audit (docling-models v2.1.0; corpus TP totals). †: the snapped-to-ground-truth count is circular by construction (Section 4). MWIS = maximum-weight independent set over the candidate graph, exact solve.

| Configuration | TP | Δ vs. raw |
|---|---|---|
| Raw (published; GT layer) | 5,204 | – |
| GT layer, snap ON† | 5,488 | +406 vs. snap-OFF |
| GT layer, snap OFF | 5,082 | −122 |
| OCR, snap ON | 3,902 | −1,019 vs. snap-OFF |
| OCR, snap OFF | 4,921 | −283 |
| OCR, snap OFF + MWIS | 5,202 | +281 vs. 4,921 |

$t(99) = 2.92$, $p = 4.4 \times 10^{-3}$), a mean paired advantage of +0.0593 TEDS-S purchased with 500 destroyed true positives (5.0 per table). Table 5 applies the pre-declared grid: among the 38 tables where the heuristic destroys at least 30% of the cells that exact selection recovers, 11 (11% of the corpus) still score ≥ 0.80 TEDS-S under the heuristic; the corresponding counts at ≥ 50% and ≥ 70% destruction are 7 of 29 and 1 of 15. Fig. 2 shows that the advantage is not coupled to the scale of destruction: if TEDS-S penalized cell loss, per-table advantages would fall below zero as destroyed cells accumulate; instead the association is weak and statistically indistinguishable from zero (Pearson $r = -0.095$, $p = 0.35$; Spearman $\rho = -0.138$, $p = 0.18$), and tables destroying 20–45 cells retain advantages up to +0.46.

Two flagship cases make the pathology concrete. In `_550868`, exact selection recovers +11 valid cells over the heuristic (36 vs. 25 TPs) yet scores 0.4566 TEDS-S lower (0.5212 vs. 0.9778). In `_597425`, the heuristic retains 4 of 102 cells—3.9%—and still scores 0.9265 TEDS-S against exact selection's 0.4748. These cases are the visible tip of the distribution in Fig. 2, not outliers: a metric that can score near-perfection on a 96%-destroyed table cannot be the sole basis for ranking TSR systems. F1 favors exact selection, whereas TEDS-S and GriTS-Top favor the destructive heuristic in this comparison (precision differs by at most 0.4 pp); we recommend that F1 and cell-level counts accompany or replace TEDS-S in post-processing studies.

Table 5: Pre-declared TEDS-S over-pruning prevalence grid (100 tables, OCR). $D$ = fraction of the cells recovered by exact selection that the heuristic destroys on that table; parenthetical shares are of the full corpus.

| Destruction $D$ | Tables | TEDS-S ≥ 0.80 | TEDS-S ≥ 0.90 |
|---|---|---|---|
| $D \geq 0.30$ | 38 | 11 (11%) | 5 (5%) |
| $D \geq 0.50$ | 29 | 7 (7%) | 3 (3%) |
| $D \geq 0.70$ | 15 | 1 (1%) | 1 (1%) |

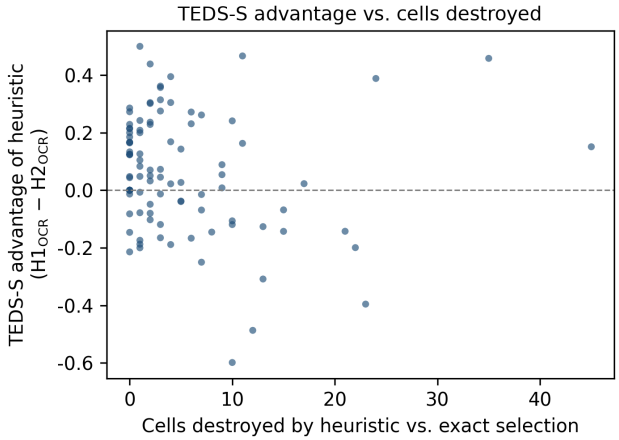



Figure 2: Per-table TEDS-S advantage of the heuristic over exact selection ($H1_{OCR} - H2_{OCR}$) against the number of ground-truth cells it destroys (100 tables, OCR). Destruction is not penalized: the advantage is unrelated to its scale (Pearson $r = -0.095$, $p = 0.35$; Spearman $\rho = -0.138$, $p = 0.18$), and tables destroying more than 20 cells retain advantages up to +0.46.

## 5. Pre-specified 1,000-table replication

All N=100 analyses above are exploratory: the protocol was fixed before the multi-metric suite ran, but no registry preregistration exists for them. The confirmatory instrument is a pre-specified study on exactly $N$=1,000 unique PubTabNet test images (spanning-tercile strata 350/350/300), drawn with NumPy `default_rng(20261016)`, replace=False, with a pinned fallback to the validation split under the same PRNG stream and pre-committed exclusion rules. Four confirmatory hypotheses are pre-committed (Holm–Bonferroni, family-wise $\alpha$=0.05): (H1a) exact selection beats the heuristic

on paired cell TPs; (H1b) the heuristic beats exact selection on paired TEDS-S (the Branch-B divergence replicates; pilot: 57/100 tables, mean +0.0593); (H2) MWIS is TOST-equivalent to raw TableFormer within ±2 TP/table; (H3) normalized greedy recovers ≥ 95% of exact ILP recall while unnormalized greedy recovers < 50%. A priori power from the pilot effect sizes is > 0.999 for H1a (mean +5.00 TP/table, SD 7.51) and the TOST margin retains > 0.999 power at the pre-specified margin; the minimum detectable effect is 0.75 TP/table at 80% power and one-sided $\alpha$=0.01. The protocol text, power computations, and pinned dependencies are part of the released artifacts; deviations, if forced by infrastructure, will be reported in the reproducibility appendix.

## 6. Limitations

Four limitations bound these findings. First, audit scale: N=100 tables with 5,646 cells permits cell-level tracing and exact statistics but not population-level estimates; the pre-specified N=1,000 replication addresses power, not breadth; both remain single-corpus (PubTabNet/PubTables-1M). Second, checkpoint provenance: the audit executes reference implementations with disclosed checkpoints (Section 3); a fully de-risked replication would re-run every system from its official release. Third, coverage of paradigms: we audit one system per paradigm; a second split-and-merge model retained the merge artifact, but we do not claim universality across the dozens of published TSR variants. Fourth, metrics: we audit TEDS-S and GriTS at their standard settings and cell TP at IoU 0.25; threshold sensitivity is included in the ledger, but other metrics (APTED variants, APD) are untested, and the OCR token condition models realistic deployment geometry without recognition errors in cell text.

## 7. Conclusion

Across four public TSR paradigms, the largest measured losses in our audit are not learned at all. A merge heuristic destroys up to a third of the cells its own split stage finds and an exact formulation restores them; a one-line threshold change near-triples a detector's usable recall; a snapped evaluation layer manufactures hundreds of true positives out of ground truth; and the field's default structure metric rewards the destruction on more than half of the audit corpus. None of these artifacts is visible in a standard leaderboard table. The audit protocol offered here—freeze the traces, replace heuristics with exact formulations under identical objectives, test equivalence as rigorously as superiority, and pre-specify the confirmatory run—is cheap (our exact solvers average 60 ms per table) and should become part of how post-processing is evaluated in table recognition and adjacent structured-prediction tasks.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Abhinav Shukla:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing. **Nachiket Tapas:** Conceptualization, Methodology, Supervision, Writing – review & editing.

## Data and code availability

All traces, per-table results, the frozen results ledger with its supersession map, exact solver code, the per-table TEDS-S prevalence analysis, and the pre-specified replication protocol are released at https://github.com/maxprogrammer007/major_project_csvtu, together with the audit-corpus construction scripts and the pinned dependency manifest (`requirements.lock`).

## References


[1] X. Zhong, E. ShafieiBavani, A. Jimeno Yepes, Image-based table recognition: data, model, and evaluation, in: Computer Vision – ECCV 2020, LNCS, vol. 12366, Springer, 2020, pp. 564–580. doi:10.1007/978-3-030-58589-1_34.

[2] C. Tensmeyer, V.I. Morariu, B. Price, S. Cohen, T. Martinez, Deep splitting and merging for table structure decomposition, in: 2019 International Conference on Document Analysis and Recognition (ICDAR), IEEE, 2019, pp. 114–121. doi:10.1109/ICDAR.2019.00027.

[3] A. Nassar, N. Livathinos, M. Lysak, P. Staar, TableFormer: table structure understanding with transformers, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 4614–4623. arXiv:2203.01017.

[4] C. Auer, M. Lysak, A. Nassar, N. Livathinos, P. Staar, et al., Docling technical report, arXiv:2408.09869 (2024). Software: DS4SD, docling-models v2.1.0, GitHub repository (2024).

[5] B. Smock, R. Pesala, R. Abraham, PubTables-1M: towards comprehensive table extraction from unstructured documents, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 4634–4642. arXiv:2110.00061.

[6] B. Smock, R. Pesala, R. Abraham, GriTS: grid table similarity metric for table structure recognition, in: Document Analysis and Recognition – ICDAR 2023, LNCS, Springer, 2023. doi:10.1007/978-3-031-41734-4_33. arXiv:2203.12555.

[7] H. Xing, F. Gao, R. Long, J. Bu, Q. Zheng, L. Li, C. Yao, Z. Yu, LORE: logical location regression network for table structure recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37(3), 2023, pp. 2992–3000. doi:10.1609/aaai.v37i3.25402.

[8] W. Lin, Z. Sun, C. Ma, M. Li, J. Wang, L. Sun, Q. Huo, TSR-Former: table structure recognition with transformers, in: Proceedings of the 30th ACM International Conference on Multimedia, 2022. arXiv:2208.04921.

[9] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.

[10] D. Lakens, Equivalence tests: a practical primer for *t* tests, correlations, and meta-analyses, Social Psychological and Personality Science 8 (4) (2017) 355–362. doi:10.1177/1948550617697177.